MODEL CARD · NATURAL LANGUAGE PROCESSING · SAII

# FINROOT: Complaint Root-Cause Classifier

*Explainable NLP for financial complaint classification and triage: TF-IDF vs MiniLM, LIME evidence, human-review gate*

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**0.70 | 0.69**
Accuracy
TF-IDF | MiniLM · majority class 0.597

**0.37 | 0.38**
Macro-F1
TF-IDF | MiniLM · weighted F1 0.65 | 0.66

**0.09 → 0.18**
Fraud recall
UNAUTHORIZED_TRANSACTION · 77 test cases

**0.591–0.807**
Confidence, one complaint
four writing styles, same wrong-by-design label

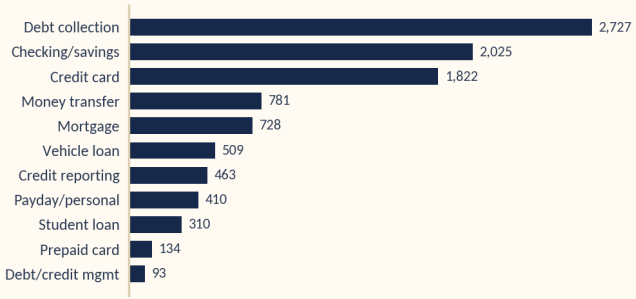
## Model details

SCIKIT-LEARN · SENTENCE-TRANSFORMERS

| | |
|---|---|
| **Task** | Multiclass root-cause prediction (10 provisional labels) from free-text complaints |
| **Pipeline** | Normalise text → classify → LIME explain → review gate |
| **TF-IDF** | 1–2-grams, 10,000 features; fast and transparent |
| **MiniLM** | all-MiniLM-L6-v2 sentence embeddings, pretrained, no fine-tuning |
| **Classifier** | Logistic Regression (max_iter 1000), same split for both; no class weights or tuning |
| **Safeguard** | Confidence below 0.60 gives "Human review required"; resolution text is hand-written, never generated |

## Training & evaluation data

CFPB · PUBLIC




- **10,002** CFPB complaint narratives (July 2026), 11 products, mean 246 words; **7,819** AI-reviewed annotation spans on 1,200 narratives.
- **Labels** are provisional (silver), not human-verified. **Split:** stratified 80/20, seed 42: 8,001 / 2,001. Post-complaint fields excluded.
- **Duplicates:** 792 flagged but not grouped in the split, so scores may be slightly optimistic.

## Intended use

IN SCOPE

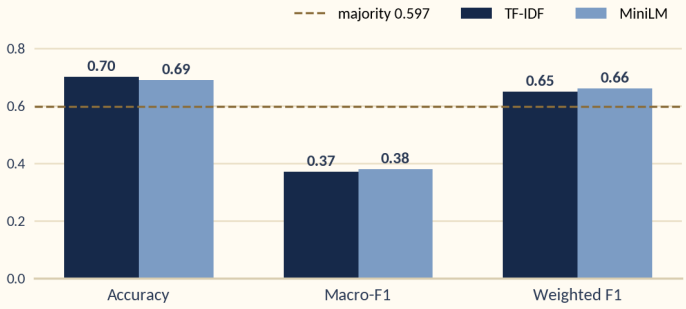
- Decision support: a likely category, a confidence, evidence words and a review decision for an analyst.
- Classroom demonstration of an explainable classify-and-triage pipeline.

OUT OF SCOPE

- Automated routing of fraud or unauthorized-activity complaints, or proving fault by a bank.
- Indian banking / UPI, Hinglish or code-mixed text; deployment without human review.

## Test-set performance

N = 2,001 · ONE SPLIT




STRONGEST CLASSES · F1 TF-IDF / MINILM

| | | | |
|---|---|---|---|
| Catch-all (1,195) | **0.81 / 0.81** | Identity theft (69) | **0.59 / 0.58** |
| Debt collection (185) | **0.56 / 0.55** | Fee / charge (123) | **0.47 / 0.50** |

Models are comparable overall, about ten points above the majority baseline. Rare classes (loan payoff, unwanted add-on) score 0.00.

## Explainability

LIME · MINILM

*"Money was deducted but receiver did not get it"*




**0.468**
predicted: OTHER_FINANCIAL_DISPUTE → human review

An explanation helps an analyst inspect a prediction; it does not prove it is correct.

## Writing-style audit

MINILM · 4 CASES

| | | |
|---|---|---|
| Formal | → same category | 0.591 |
| Informal | → same category | 0.726 |
| Noisy | → same category | 0.668 |
| Hinglish | → same category | 0.807 |

FAILED_TRANSFER is not a trained label, so these cases show behaviour only: confidence peaked on the input the model cannot read. Not a fairness study.

## Key findings

- **Collapse to the catch-all:** 61 of 77 unauthorized-transaction complaints (79%) were predicted as OTHER_FINANCIAL_DISPUTE.
- **Confidence is not a trust signal:** it rose highest on Hinglish, so it is paired with explanations and review.
- **Embeddings:** raise fraud recall (0.09 → 0.18) but give no clear overall gain.

## Limitations & risks

- **Silver labels:** scores measure agreement with provisional labels.
- **Resolution gap:** 2 hand-written procedures, neither matching a classifier label, so every prediction ends in human review.
- **Rerun on 288 labelled rows** (230 / 58): TF-IDF 0.66 / 0.24, MiniLM 0.71 / 0.35 vs 0.534 majority; too small to rank models.
- Uncalibrated 0.60 gate; U.S. English only; MiniLM truncates long text.

## Recommendations

- **Decision support only,** with human review for low confidence and fraud text.
- Fix data loading, then re-run with cross-validation and confidence intervals.
- Human-verify labels; split OTHER; add class weights; dedupe before splitting.
- Link resolutions to labels; add UPI and Hinglish data.

**Bottom line:** data and label quality matter more than model choice.